

Classification of Left/Right Hand Movement from EEG Signal by Empirical Mode Decomposition with Ensemble of classifiers

Neeraj Choudhary

Dept. of Computer Science and Engg.

Mahindra University

Hyderabad-500050, India

neeraj.choudhary@mahindrauniversity.edu.in

Kodidasu Murali Kumar

Dept. of Computer Science and Engg.

Indian Institute of Technology Patna

Patna-801103, India

muralikumarkodidasu@gmail.com

Jimson Mathew

Dept. of Computer Science and Engg.

Indian Institute of Technology Patna

Patna-801103, India

jimson@iitp.ac.in

Sravanthi Achchugatla

Dept. of Computer Science and Engg.

Mahindra University

Hyderabad 500090, India

sravanthi.achchugatla@mahindrauniversity@edu.in

Abstract—Electroencephalography (EEG) is a brain wave signal that plays a significant role in the neural control interface. EEG signals have many nonlinear dynamic properties; because of these properties, handling EEG signals for modeling with standard linear approaches will lead to inconsistent results. EEG has a significant importance in the brain-computer interface (BCI). BCI has demonstrated a great deal of potential to benefit the human way of life. Additionally, its application targets people with disabilities. BCI extracts the informative features from acquired brain signals and generates signals knowing these features, which can control external devices. EEG signals can show improvement in classifying motor imagery tasks. In this paper, we introduce a new method for extracting time-frequency-related properties on different scales using empirical mode decomposition (EMD) and classifying the left and right hand movements of the subject. EMD is a signal processing technique that decomposes a signal into different frequency bands of the same length. We use an ensemble of classifiers with bootstrapped data for classification. The proposed method performs better than the previous method [1] in terms of accuracy and F1-score. The empirical results show that the proposed method has an average accuracy of 97.30% and an average F1 score of 96.35%.

Keywords—Empirical mode decomposition, ensemble of estimators.

I . INTRODUCTION AND LITERATURE REVIEW

Electroencephalography (EEG) is a visual representation of the electrical activity of neurons. A medical student, John Larson, in 1921, invented the modern polygraph instrument for deception detection. Later, Berger, in 1929, developed electroencephalography, a graphical representation of the contrast in voltage between two different cerebral locations plotted over time. He described the human alpha and beta rhythms. EEG is a scalp activity that we record from the brain using electrodes. There are different operators to record it. There are over a billion neurons in the brain that communicate with each other through electrochemical signals. The changes in these electrodes, which are recorded using scalp electrodes as

continuous changes in voltages over time, are EEG. There is one amplifier between the pair of electrodes. These amplifiers amplify the voltage difference between two electrodes. Single-neuron activity is too small to be detected by EEG. These EEG signals are weak and susceptible to noise. Generally, recording EEG signals is non-invasive and involves simply placing electrodes on the scalp. EEG is relatively cheap compared to neuro-imaging techniques. EEG signals can be used in diagnostic applications that focus on event-related potentials and the spectral content of EEG. There are other medical uses, like the detection of epilepsy, dementia, hemorrhage, cognitive science, and seizures, etc. EEG has many game development and smart home application implications [2][13]. People with physical disabilities, such as those with motor disabilities, can use EEG signals to express their feelings, emotions, and intentions. There are other types of brain signals, but when compared with EEG signals, other signals have a high signal-to-noise ratio. Patterns of the EEG will change based on the subject's state, like sleeping, focusing, concentrating, etc. There are different types of wave patterns in EEG based on different frequency ranges. Delta [0.5–4 Hz] is related to slow sleep in adults. Theta [4–8 Hz] is related to drowsiness in children and adults. Alpha [8–14 Hz] is related to the closing of eyes, relaxation, and attenuation with eye opening or mental exertion. Mu [8–13 Hz] is related to resting state motor neurons; Beta [14–30 Hz] is related to active, busy, or anxious thinking and active concentration. Including motor imagery movements, EEG signals also have the property of having the capacity to contain memory-related effects. Fig.1 depicts the different brain waves and their range of frequencies.

Analyzing non-stationary and non-linear time series is done using the Hilbert transform and empirical mode decomposition

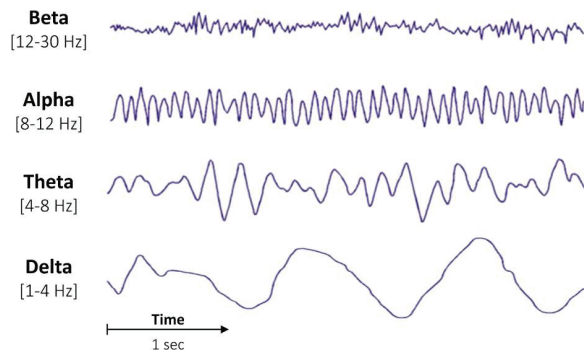


Fig. 1: Different brain waves

[3]. As empirical model decomposition decomposes any data into finite and smaller intrinsic mode functions, As it is an adaptive method, it is highly effective and efficient. It is very applicable to nonlinear and non-stationary data because the decomposition is based on the local characteristic time scales of the given data. The Hilbert transform generates immediate frequencies as functions of time, which allows recognition of embedded patterns. The effectiveness of intrinsic mode functions makes the instantaneous frequency significant. This instantaneous frequency can remove the fake harmonics and constitute a nonlinear and non-stationary signal.

The Hilbert-Hung transform is preferable for extracting factual elements that are peculiar to fundamental biological or physiological processes [4]. As the process is adaptive, it generates complete orthogonal basis functions. Hilbert-Hung consists of empirical mode decomposition with the Hilbert transform. The property of rendering over the time of IMFs makes them dissimilar from traditional basis systems and perfectly suited to analyzing non-stationary and nonlinear time series.

EEG signal analysis and classifying familiarity and unfamiliarity will help in deception detection [5][12]. In this paper, the author attempted to classify the familiarity and unfamiliarity of EEG signals using independent component analysis and an autoencoder. They acquired the data set using an 8-electrode OpenBCI Ultracortex "Mark IV" EEG headset to capture EEG signals. For classification, they used O1, O2, P7, and P8 electrodes, which are related to the visual cortex. Independent component analysis can separate the components in the EEG. The non-linear feature extraction property of the autoencoder helped to get better accuracy.

Self-organization maps (SOM) are a neural network technique that can classify the left- or right-hand movement of an EEG signal [1], 2014 classification). The MAP in SOM tries to trace the input by configuring the coefficients. The nodes of SOM develop themselves, the learning capacity of SOM depends on how little the change is. It is an unsupervised technique.

In [6], wavelet transform for classification of the left-hand and right-hand movements of EEG signals Wavelets

help to explore time domain information. The Morlet wavelet transform helped analyze the alpha and beta brain waves and extract time-domain features. The feature subset selection algorithm helps reduce the feature set. A neural network with a multilayer back propagation network classifies the signals by extracting nonlinear properties from the data. The remainder of this paper is organized as follows: Section 2 follows the data description and processing. Section 3 follows with a brief summary of the methods used. Section 4 discusses the results, and Section 5 is an outlook on the conclusion.

I. DATA DESCRIPTION AND PREPROCESSING

In this section, a detailed data description and data preprocessing technique are discussed. We have used EMD to divide the data into different frequency bands. The EMD and EMD algorithm are described in the section that follows.

A. Empirical Mode Decomposition

Empirical mode decomposition (EMD) is decomposing the signal without leaving the time domain [7] [3][13]. It assumes that the time series non-stationary and nonlinear contains distinct essential modes of oscillation. The quintessence of this method is that it identifies these essential oscillatory modes empirically with the help of their characteristic time scales in the data and decomposes the data accordingly [8]. EMD decomposes the signal into a number of frequency modulated and amplitude-modulated zero mean signals. The decomposed IMFs follow the following properties.

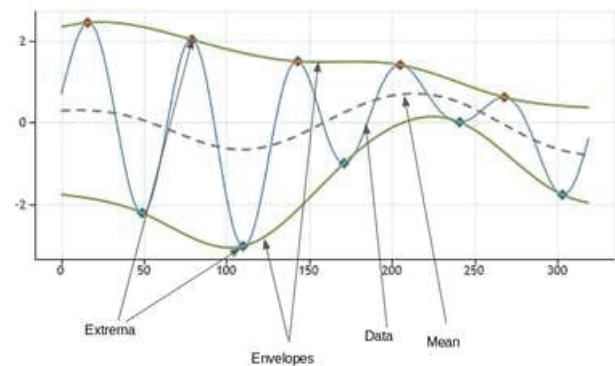


Fig. 2: Signal Localization

- Every IMF is the difference between a number of extrema and zero crossings either zero or one.
- At any point in IMF the mean value is zero.

B. EMD Algorithm

The algorithm steps are as follows

- 1) Find the all extrema (maxima and minima) in the given data $X(t)$.
- 2) By using cubic spline interpolation form the envelopes based on extrema points.
- 3) Form a mean function $M1(t)$ of top and bottom envelopes of the data.

- 4) Calculate $D(t) = X(t) - M1(t)$, if $D1(t)$ is zero mean then accept the $D1(t)$ as first IMF ie $H1(t)$
- 5) Else repeat the step 1 to step 4 by considering $D(t)$ as data.

Fig. 2 shows the signal localization with extrema, envelopes, data, and mean points. Once the first IMF is obtained we will get the remaining IMFs for the residual signal. The residual signal contains components of lower frequency. $R(t) = X(t) - H1(t)$. We can get the number of IMFs till the residual signal has at most one extrema or simple monotonic signal. The output of EMD forms almost orthogonal basis functions to the original signal [9]. These decomposed IMFs have the same length as the original signal and in the time domain. All the IMFs are non-overlapped.

There is a public EEG data set BCI competition IV data set IIb. This data is collected from nine subjects, and the data is recorded with the help of three bipolar electrodes C_2 , C_3 , C_4 . The data was collected at a sample rate of 250HZ.

These signals are filtered between 0.5HZ and 100HZ with the help of bandpass filtering. This recording happens with the help of cue-based patterns of motor imagery tasks of the left hand(class 1) and right hand(class 2). This data set contains a total of five sessions for each subject, first three sessions contain the class label and the last two sessions are for testing so it don't have class labels.

C. Preprocessing

In this experiment, we used the first three data sets for training and testing. Initially, we used a bandpass filter to filter signals between 4 and 40 HZ.

Now the output signal is decomposed into IMFs using the EMD algorithm. For each channel, we have decomposed IMFs separately. EMD decomposed each channel into the number of IMFs. We calculated the correlation coefficient for each IMF and the original signal. We chose the IMF, whose correlation coefficient is greater than 0.1. Fig. 3 shows the original signal along with the first four useful IMFs. The obtained IMFs are used as features for classification. The original signal is the raw data or the signal you start with. It could be any time-series data or a signal that you want to analyze. The first Intrinsic Mode Function (IMF) is obtained through a process called Empirical Mode Decomposition (EMD). EMD decomposes a signal into components called IMFs, which represent oscillatory modes. The first IMF typically captures the most dominant and rapidly varying pattern in the original signal. The second IMF represents another level of the signal's decomposition. It captures the next level of oscillatory behavior that is orthogonal (independent) to the first IMF. The third IMF captures another level of oscillatory behavior, orthogonal to the first and second IMFs. This continues the decomposition process, capturing additional oscillatory patterns in the signal.

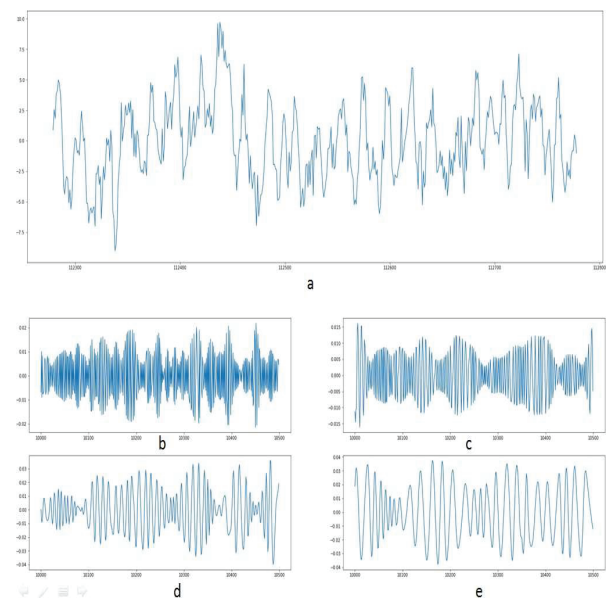


Fig. 3: (a) original signal (b) first IMF (c) second IMF (d) third IMF (e) fourth IMF

II. METHODOLOGY

The proposed approach is an ensemble of decision trees with bootstrapping as a classifier. Bootstrapping helps ensemble learners; we can build them using the same algorithm but train each learner on distinct sets of data, this is also called bagging. First, we have to create the bootstrap data, for this, we have to create m bags for m data sets. Then, from the training data, we will add n points to the first bag. These n points will be taken at random from the training data with replacement. Fig.4 is the visual representation of bagging, n data points collected in each bag from training data. Now we have created m bags

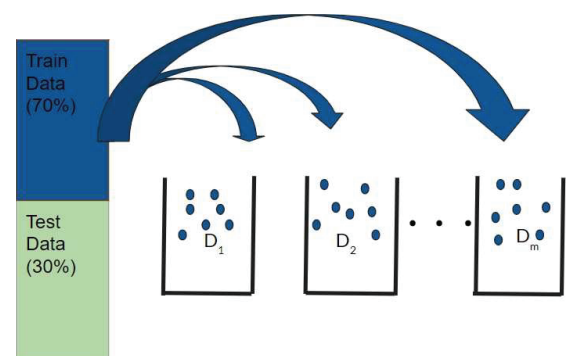


Fig. 4: Bootstrapping data sets

with n data instances chosen at random with replacement. In general, n is 60% of the actual training data set. Now we will build m distinct models, each one for each different data set. We will build m different models by randomly selecting different variables at distinct times. Here we are building a decision tree as a model. The steps to build this ensemble decision tree are as follows:

- Use the bootstrapped data set.
- Select two random variables at every step.

Fig.5 is the m trees randomly built from m different data sets constructed with bootstrapping. Now every decision tree

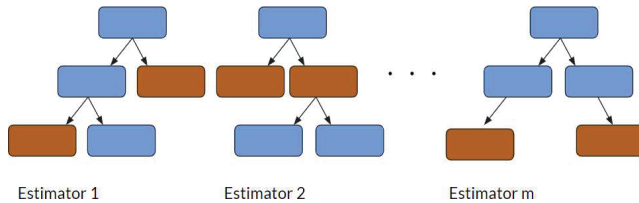


Fig. 5: Decision trees created from m different data sets

will give its output, in ensembles we have to combine the results of all the estimators and take the average. Bootstrapping the data and aggregating it to make decisions is called bagging. Random forests are supervised learning models [10]. The decision tree splits the working data into smaller sub-parts repeatedly. It uses simple if-then-else decision rules. If the depth of the tree is greater, then the model and rules are more complex. We used the Gini index to build the decision tree. The Gini index works on the rule that if the sample is pure, then selecting two items from the sample must be of the same class and the probability for this is one [11]. It executes only the binary split; if homogeneity is higher, then the Gini is higher.

III. RESULTS

We tested with a different number of estimators and at different level depths. When the number of estimators increases and the depth also increases, then the model is giving good performance. But when the depth of the tree is decreasing, the classification performance is poor. The Fig.6

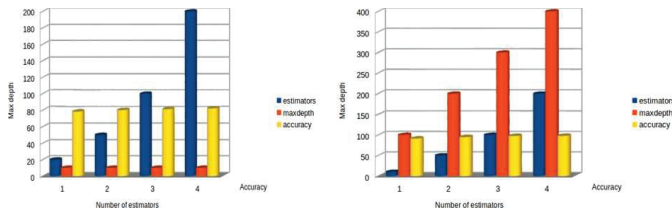


Fig. 6: Accuracy with respect to depth and number of estimators.

Describes how the accuracy is changing based on the number of estimators and the maximum depth of the tree.

Accuracy and F1-Score		
Subject	Accuracy	F1-score
Subject 1	97.82	97.41
Subject 2	97.24	96.17
Subject 3	98.12	98.24
Subject 4	96.22	96.04
Subject 5	96.15	95.81
Subject 6	97.88	95.31
Subject 7	96.93	97.12
Subject 8	97.73	95.86
Subject 9	97.52	95.31

TABLE I: Accuracy and F1-Score

We also tried without bootstrapped data, in this case, we built the number of estimators with the same data set. But when we made the estimator, we selected variables randomly. This model gives 94% accuracy overall. The parameters and number of variables to select need to be tuned.

We applied the ensemble of decision trees with bootstrapped data sets for each person separately. Table I shows the results in terms of accuracy and F1-score. The classifier gives an average accuracy of 97.30% and an F1-score of 96.35%. We also calculate the F1-score by finding precision and recall for each subject separately. These results clearly show that the model's performance is better than previous models.

IV. CONCLUSION

We present an efficient approach for classifying motor imagination on EEG signals using EMD. Compared with the methods used in [7] [3], EMD is a technique for adaptively representing nonstationary signals and instantaneous frequency for the removal of fake harmonics. Utilizing time-domain analysis in frequency bands, the output of the EMD is divided into IMFs that are the same length as the original signal. We used an ensemble of decision trees with the bootstrapping bagging method. The model worked well when we utilized the Gini Index discussed in the methodology introduced in the case study [11] to compare the behavior of the Gini Index and Information Gain criteria to assess different decision tree level depths. We used a decision tree classifier to categorize the data, and on average, we got 97.30% accuracy and 96.35% F1 score compared with the method used in [1] where the accuracy is 86%.

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